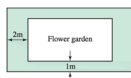


Problem 2

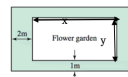
Problem 2

Problem 1 A rectangular flower garden with an area of 30 m^2 is surrounded by a grass border 1 m wide on two sides and 2 m wide on the other two sides (see figure). What dimensions of the garden minimize the combined area of the garden and borders?



- (a) Label the picture with variables.

Problem 2



Explanation.

- (b) *What are you trying to maximize or minimize? Write an equation for it in terms of the variables from (a).*

Explanation. We want to minimize the combined area of the garden and border. If A denotes this area, then an equation for A is

$$A = (x + 4)(y + 2)$$

- (c) *What is your constraint? Write a constraint equation in terms of the variables from (a).*

Explanation. We know that the area of the flower garden is 30 m^2 . So our constraint equation is

$$xy = 30$$

- (d) *Reduce your optimization equation to one variable using the constraint equation.*

Explanation.

$$x = \frac{30}{y} \quad (\text{Note that } y \neq 0)$$

$$A = \left(\frac{30}{y} + 4 \right) (y + 2) \quad (1)$$

$$= 30 + 4y + \frac{60}{y} + 8 \quad (2)$$

$$= 4y + \frac{60}{y} + 38 \quad (3)$$

- (e) What is the interval on which your variable makes sense? Is it open or closed? What does this mean for the method of finding the global max or min?

Explanation. $0 < y < \infty$, which is an open interval. So we need there to only be one critical point to equation (3) in the domain of y , and then we need to show that this critical point is a local minimum. This will imply that the critical point is a global minimum (since there is only one local extremum).

- (f) Use the appropriate method to find and justify your global extremum.

Explanation. We need to differentiate equation (3) with respect to y , set this derivative equal to 0, and then solve:

$$\frac{dA}{dy} = 4 - \frac{60}{y^2} := 0$$

$$\frac{60}{y^2} = 4$$

$$4y^2 = 60$$

$$y^2 = 15$$

$$y = \pm\sqrt{15}$$

Since $-\sqrt{15}$ is not in our domain, the only critical point of $A(y)$ in the interval $(0, \infty)$ is $\sqrt{15}$. Thus, if $y = \sqrt{15}$ is a local minimum for A , then it will be a global minimum. Using the second derivative test:

$$\frac{d^2A}{dy^2} = \frac{120}{y^3}$$

$$\left[\frac{d^2A}{dy^2} \right]_{y=\sqrt{15}} = \frac{120}{15\sqrt{15}} > 0.$$

Since $A(y)$ is concave up at $y = \sqrt{15}$, this point is a local (and thus, global) minimum of $A(y)$. Note that we also could have used the first derivative test to show that $y = \sqrt{15}$ was a local minimum of A .

- (g) Be sure to answer the question asked in the original problem.

Explanation. Since $y = \sqrt{15}m$, we have that

$$x = \frac{30}{y} = \frac{30}{\sqrt{15}} = \frac{30\sqrt{15}}{15} = 2\sqrt{15}m$$